

DATABASE MANAGEMENT SYSTEMS LAB

II B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS10	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
COURSE OBJECTIVES								
The course should enable the students to:								
1. Apply the basic concepts of Database Systems and Applications. 2. Use the basics of SQL and construct queries using SQL in database creation and interaction 3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system. 4. Analyze and Select storage and recovery techniques of database system.								
COURSE OUTCOMES								
The course should enable the students to:								
1. Apply the basic concepts of Database Systems and Applications. 2. Develop an ER model for a given database. 3. Use the basics of SQL and construct queries using SQL in database creation and interaction. 4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system. 5. Analyze and Select storage and recovery techniques of database system.								
LIST OF EXPERIMENTS								
WEEK - 1	DDL COMMANDS							
Creation of Tables using SQL-Overview of using SQL tool and Data types in SQL a) Altering Tables b) Dropping Tables								
WEEK - 2	CREATE TABLE WITH PRIMARY KEY AND FOREIGN KEY & DML COMMANDS							
Creating Tables(along with Primary and Foreign keys),Practicing DML commands- a) Insert b) Update c) Delete								
WEEK - 3	SELECTION QUERIES							
Practicing Select command using following operations a) AND,OR b) ORDERBY c) BETWEEN d) LIKE e) Apply CHECK constraint								
WEEK - 4	AGGREGATE FUNCTIONS AND VIEWS							
Practice Queries using following functions a) COUNT b) SUM c) AVG d) MAX e) MIN Apply constraint on aggregation using a) GROUPBY b) HAVING VIEWS Create, Modify and Drop								
WEEK - 5	NESTED QUERIES							
Practicing Nested Queries using UNION, a) INTERSECT b) CONSTRAINTS c) IN								

WEEK - 6	CO-RELATED NESTED QUERIES
Practicing Co – Related Nested Queries using a) EXISTS b) NOT EXISTS c) ANY d) ALL	
WEEK - 7	JOIN QUERIES
Practicing Join Queries using a) Inner join b) Outer join c) Equi join d) Natural join	
WEEK - 8	TRIGGERS
Practicing Triggers Creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
WEEK - 9	PROCEDURES
Procedures -Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
WEEK - 10	CURSORS
Cursors -Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor	
WEEK - 11	PL/SQL Part 1
Practice PL/SQL a) Block structure b) Variables c) data types	
WEEK - 12	PL/SQL Part 2
Practice PL/SQL a) Operators b) Control structures Case study 1: College Management Case study 2: An Enterprise/Organization Case study 3 : Library Management system Case study 4: Sailors and shipment	
TEXT BOOKS	
1. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill, 3rd Edition 2. Database System Concepts, Silberschatz, Korth, McGraw Hill, 5th edition.	
REFERENCE BOOKS	
1. Database Systems design, Implementation, and Management, PeterRob & Carlos Coronel 7 th Edition. 2. Fundamentals of Database Systems, ElmasriNavrate, Pearson Education 3. Introduction to Database Systems, C.J.Date, Pearson Education 4. Oracle for Professionals, The XTeam, S.Shah and V.Shah, SPD. 5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI. 6. Fundamentals of Database Management Systems, M.L. Gillenson,Wiley Student Edition.	